Civil Engineering Department
National Institute of Technology Kurukshetra

Name of Examination: B. Tech. (Civil) III Semester Ist Mid-sem Exam
Time: **50 min** Max. Marks: **45**
Note: Attempt all questions.

1. Derive expressions for (**any two**)
   i. Capillary rise in a small diameter tube
   ii. Pressure inside a hemispherical drop
   iii. Shear stress in a viscous fluid **(4)**/4 min

2. Define and distinguish between (**any two**) following set of fluid properties:
   i. Newtonian and Non Newtonian Fluids
   ii. Forces of Cohesion and adhesion
   iii. Path line and Streak line **(4)**/4 min

3. Explain with mathematical expression (**any two**)
   i. Steady flow and unsteady flow
   ii. Uniform and non-uniform flow
   iii. Laminar flow and Turbulent flow **(4)**/4 min

4. A rectangular plate 0.50 x 0.50 m weighing 500 N slides down an inclined plane 30° with the horizontal at a velocity of 1.75 m/sec. If the 2mm gap between the plates is filled with oil, find the viscosity of the oil in poise and in Ns/m^2. **(6)**/5 min

5. For the following velocity vector determine the magnitude of velocity at A (x = 2, y = -3, z = 1, t = 2)
   $V = (10t + xy)i + (-yz - 10t)j + (-yz + z^2/2)k$
   where i, j, k are unit vectors. **(6)**/6 min

6. Find the velocity potential if the velocity field is given as
   $u = x/(x^2 + y^2)$ and $v = y/(x^2 + y^2)$ **(6)**/6 min

7. State the basic principle on which equation of continuity is based and then derive the Continuity equation in differential form. **(6)**/6 min

8. Derive an expression for total pressure and centre of pressure for a plane lamina immersed in a fluid in a vertical orientation. **(6)**/6 min

9. What do you understand by a manometer? Explain. **(3)**/3 min

THE END